# Multi-Agent NPC Simulation: System Architecture and Technical Documentation

A Hybrid AI Framework Combining Classical Decision Algorithms with Grounded Large Language Model Dialogue

**Semester AI Project Team**

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**Abstract**

Traditional video game Non-Player Characters (NPCs) typically suffer from either rigid, scripted predictability or unconstrained language model hallucinations that violate game rules and spatial reality. This report presents the architectural design and technical specification of a small-scale multi-agent simulation in which 5 to 10 autonomous game characters perceive a dynamic, discrete town environment, maintain private episodic memory streams, and execute actions governed by classical Artificial Intelligence algorithms. Classical algorithms—including $A^*$ search for deterministic pathfinding, Genetic Algorithms (GA) for multi-objective daily planning, Multi-Layer Perceptrons (MLP) for reflex action classification, and K-Means for unsupervised behavior analysis—manage physical navigation, resource allocation, and physics verification. Concurrently, a Large Language Model (LLM) functions strictly as a conversational brain, generating persona-consistent dialogue grounded solely in the NPC's private memory log. By establishing the simulation engine as the immutable ground truth, this hybrid framework eliminates state hallucinations while preserving emergent social dynamics and natural language interaction.

**Keywords:** Multi-Agent Systems, Non-Player Characters, Classical AI, Genetic Algorithms, Grounded LLM, $A^*$ Pathfinding, Game Simulation.

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# 1 Project Overview

## 1.1 Problem Formulation and Motivation

In interactive entertainment and synthetic simulations, Non-Player Characters (NPCs) have historically relied on Finite State Machines (FSMs) or static behavior trees. While computationally trivial and deterministic, these characters display zero episodic recall, fail to adapt to emergent events, and exhibit repetitive conversational dialogue.

Recent advancements in generative artificial intelligence have prompted attempts to replace scripted logic entirely with Large Language Models (LLMs). However, pure LLM-driven agents introduce critical architectural flaws:

1. **Factual and Physical Hallucination:** Generative models invent unearned items, claim impossible physical actions, or bypass geographic barriers.
2. **Loss of Spatial Grounding:** Language models lack internal metric geometry and cannot reliably compute shortest paths or avoid collision zones without structured external engines.
3. **Excessive Computational Overhead:** Querying frontier models for every low-level movement tick is computationally prohibitive and latency-heavy.

To resolve this dichotomy, we propose a **Hybrid Multi-Agent Simulation**. The system populates a simulated town (comprising nodes such as Market, Café, Town Square, Library, Park, and Residential Homes) with 5 to 10 autonomous NPCs (e.g., Ahmed, Sara, Bilal). Each character maintains sensory constraints, an individual memory stream, and dynamic goal hierarchies, arbitrated through a clear division of responsibilities (Table 1).

Table 1: Architectural Division of Responsibilities Between AI Subsystems

| Subsystem | Core Responsibilities and Guarantees |
|---|---|
| **Classical AI Engine** (*"The Body"*) | • Deterministic spatial traversal across town graph topologies ($A^*$, BFS).<br>• Multi-objective combinatorial route optimization (Genetic Algorithm).<br>• Instantaneous state-to-action reflex classification (Neural Net MLP).<br>• Unsupervised clustering of agent trajectories into emergent styles.<br>• Verification of physical laws, inventory transfer, and clock ticks. |
| **LLM Dialogue Brain** (*"The Voice"*) | • In-character natural language communication with players and NPCs.<br>• Bounded conversational replies conditioned strictly on private memories.<br>• High-level semantic intention parsing and goal decomposition. |

## 1.2 Academic Syllabus Integration

Rather than reducing AI coursework to prompt engineering, this project centers on classical and modern foundational AI pillars:

- **Graph Search ($A^*$, BFS):** Optimal point-to-point traversal across weighted topological maps.
- **Evolutionary Computation (GA):** Chromosome encoding, crossover, mutation, and elitism for multi-objective schedule planning.
- **Supervised Learning (ANN / MLP):** Feature extraction, forward propagation, loss optimization, and evaluation (confusion matrices, F1-scores).
- **Unsupervised Learning (K-Means):** Grouping high-dimensional agent trajectories to uncover emergent archetypes.

- **Grounded Generative AI:** Structured JSON prompt synthesis ensuring grounded, hallucination-free dialogue.

# 2 Inspiration and Related Work

## 2.1 Generative Agents (Park et al., 2023)

Park et al. [1] demonstrated that LLM-driven agents could exhibit coherent social behaviors in a 25-character virtual community ("Smallville"). Their core contribution was the **Memory Stream** architecture: agents record raw natural language observations, retrieve memories via recency, importance, and relevance metrics, and periodically synthesize higher-level reflections. While groundbreaking, Generative Agents relied on unconstrained LLM calls for physical planning, incurring severe compute costs and occasional spatial inconsistencies. Our architecture preserves the memory stream concept while transferring physical planning to deterministic classical algorithms.

## 2.2 Narrative Evaluation and NARRA-Gym (Huang et al., 2026)

Huang et al. [2] benchmarked nine leading frontier LLMs in multi-turn interactive storytelling environments. Their findings revealed that even state-of-the-art models consistently fail at long-horizon consistency, character stability, and spatial bounds when operating without external structured harnesses. Decoupling the narrative dialogue layer from the physical simulator directly resolves these stability deficits.

## 2.3 Text-Based Environments and Gym Standards

Microsoft's TextWorld [3] established that discrete room environments and text interfaces provide rigorous, reproducible benchmarks for reinforcement learning and planning. Concurrently, the Gymnasium [4] and PettingZoo [5] (Farama Foundation) standards formalize multi-agent simulation loops ('reset()', 'step()', observation/action spaces). Our town engine adopts Gymnasium conventions, ensuring lightweight execution on standard multi-core CPUs without GPU dependencies.

# 3 System Architecture and the Simulation Loop

The simulation operates as an episodic, discrete-time loop where each tick represents a fixed quantum of virtual time (e.g., 5 simulation minutes), as illustrated in Figure 1.

## 3.1 Detailed Stages of the Simulation Tick

### 3.1.1 Stage 1: World State (Ground Truth)

The central game engine maintains the objective ground truth of the universe:

$$S_t = \langle \mathcal{G}, \mathcal{O}_t, \mathcal{A}_t, \tau_t \rangle \tag{1}$$

where $\mathcal{G} = (V, E)$ represents the town graph topology, $\mathcal{O}_t$ denotes item placements and states, $\mathcal{A}_t$ represents character locations and inventories, and $\tau_t$ tracks the simulation clock.

### 3.1.2 Stage 2: Perception Filtering

To prevent omniscient knowledge leakage, each agent $i$ receives a filtered observation $o_{i,t} \subset S_t$ constrained by a spatial sensory radius $r_{\text{percept}}$:

$$o_{i,t} = \{e \in S_t \mid \text{dist}(\text{loc}(i), \text{loc}(e)) \leq r_{\text{percept}}\} \tag{2}$$

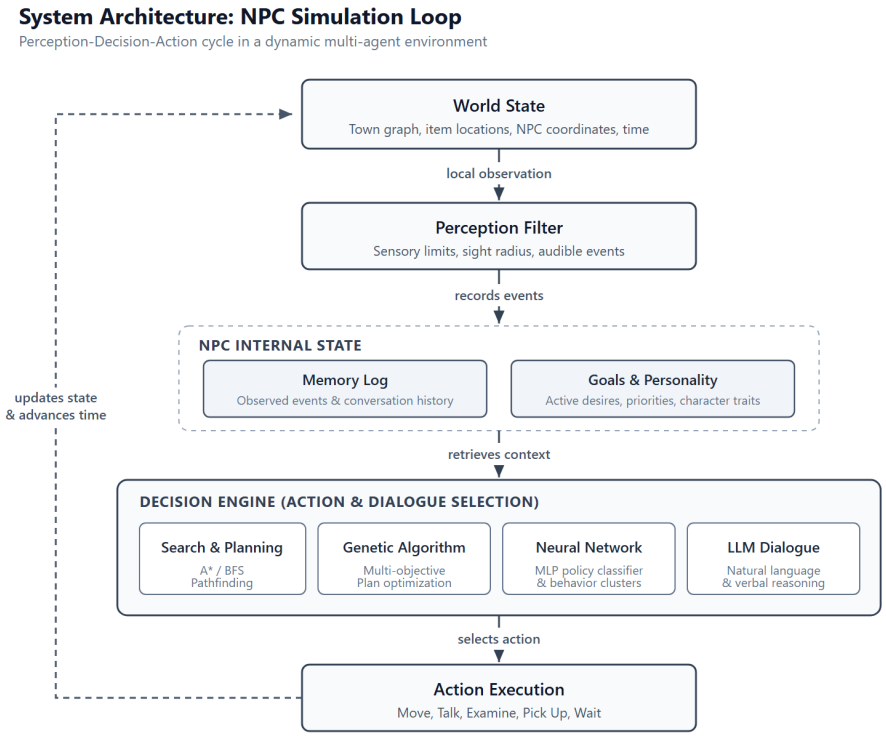



Figure 1: The closed perception-decision-action simulation loop governing agent behaviors.

Events occurring beyond this sensory boundary are withheld from the character's cognitive pipeline.

#### 3.1.3 Stage 3: Internal NPC State Representation

Each character maintains two persistent internal structures:

1. **Episodic Memory Log ($M_i$):** An append-only chronological ledger of observations, dialogues, and interactions.
2. **Goal and Trait Profile ($P_i$):** A static vector of personality traits (e.g., caution $\in [0, 1]$, curiosity $\in [0, 1]$) and a priority stack of active operational goals.

#### 3.1.4 Stage 4: Hybrid Decision Engine

Given internal state $\langle M_i, P_i \rangle$ and current observation $o_{i,t}$, the character evaluates whether physical navigation, multi-stop planning, reflexive action, or dialogue is required, dispatching the task to the respective algorithm.

#### 3.1.5 Stage 5: Action Execution and State Validation

The proposed candidate action $a_i \in \{\text{Move}, \text{Talk}, \text{Examine}, \text{PickUp}, \text{Wait}\}$ is submitted to the simulation engine. The engine verifies physical feasibility (e.g., target object adjacency, path validity). Valid actions mutate the world state $S_{t+1}$ and advance the clock $\tau_{t+1}$.

**The Grounding Axiom:** World State is absolute truth. The language model may formulate dialogue and intent, but cannot modify physical coordinates, alter item inventories, or warp simulation time.

# 4 The AI Decision Engine: Classical AI and LLM Integration

The decision engine routes computational subtasks to specialized algorithms, ensuring verifiable execution and linguistic richness (Figure 2).

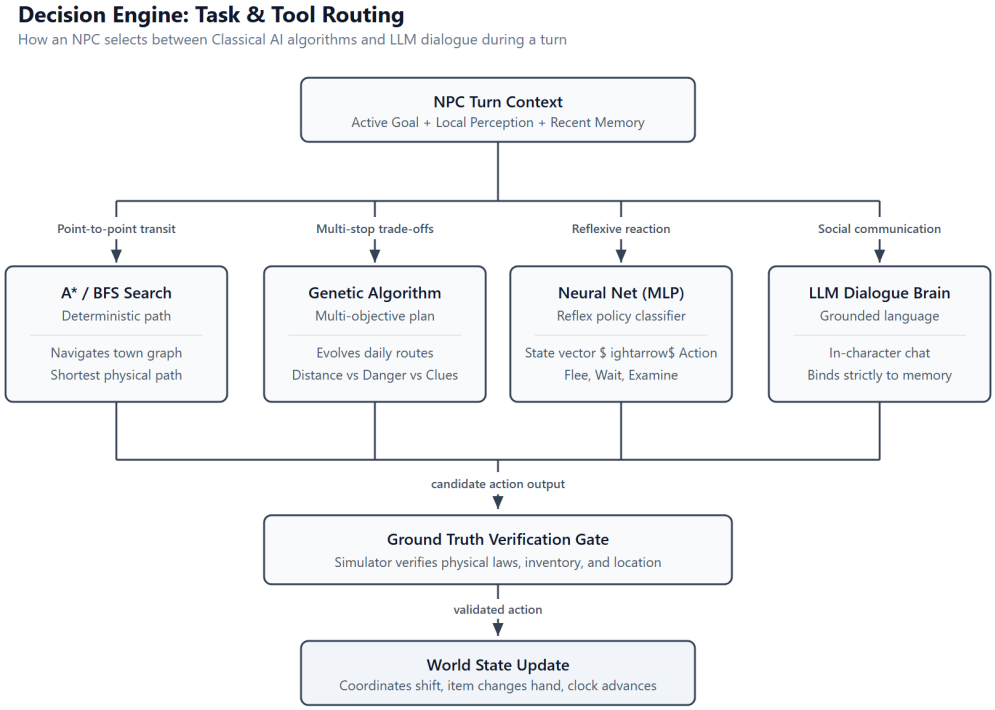



Figure 2: Task and tool routing within the hybrid decision engine.

## 4.1 Search and Pathfinding ($A^*$ and BFS)

When an agent requires traversal between two distinct town nodes (e.g., Home to Market), the system invokes the $A^*$ search algorithm across the weighted town graph $G = (V, E)$. The evaluation function:

$$f(n) = g(n) + h(n) \tag{3}$$

combines exact path cost $g(n)$ from the origin with an admissible Euclidean distance heuristic $h(n)$ to the destination node. This guarantees mathematically optimal paths with zero computational hallucination.

## 4.2 Genetic Algorithm for Multi-Objective Planning

When an NPC must construct a daily itinerary balancing competing requirements, single-source shortest path algorithms fail. We formulate multi-objective daily scheduling as an evolutionary optimization problem (Figure 3).

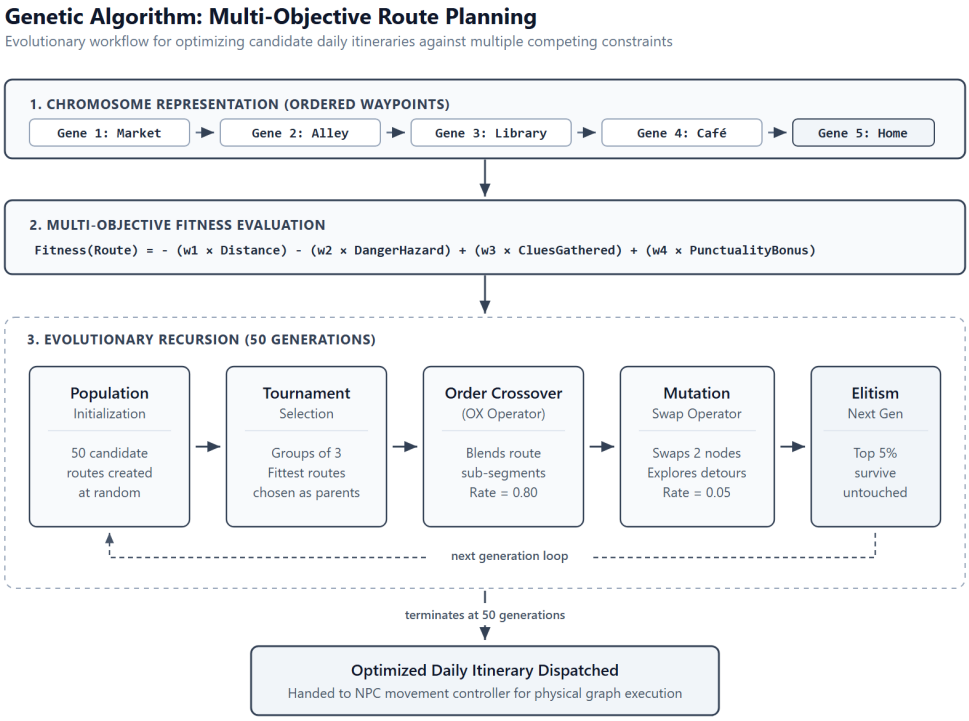



Figure 3: Genetic algorithm chromosome structure, fitness evaluation, and evolutionary cycle.

### 4.2.1 Chromosome Representation

A candidate plan is encoded as an ordered permutation chromosome of target waypoint nodes:

$$C = [v_1, v_2, v_3, \ldots, v_k], \quad v_j \in V \tag{4}$$

For example: [Market → Alley → Library → Café → Home].

### 4.2.2 Multi-Objective Fitness Evaluation

The fitness function penalizes distance and danger while rewarding clue discovery and punctual completion:

$$\text{Fitness}(C) = -(w_1 \cdot D(C)) - (w_2 \cdot H(C)) + (w_3 \cdot R(C)) + (w_4 \cdot B(C)) \tag{5}$$

where $D(C)$ represents cumulative travel distance, $H(C)$ denotes passage through high-threat alleys, $R(C)$ quantifies investigated clue sites, and $B(C)$ is a terminal punctuality bonus.

### 4.2.3 Evolutionary Operators

- **Initialization:** 50 candidate routes generated randomly from valid graph connections.
- **Tournament Selection:** Subsets of $k = 3$ individuals compete; the highest fitness candidate is selected as a parent.
- **Order Crossover (OX):** Preserves contiguous route sub-segments while preventing duplicate node visitation.
- **Swap Mutation:** Randomly swaps two waypoints ($p_m = 0.05$) to explore novel detours.

- **Elitism:** The top 5% fittest candidates survive into subsequent generations without mutation.

The algorithm terminates after 50 generations (taking $< 200$ ms on CPU), dispatching the evolved itinerary to the agent controller.

### 4.3 Neural Network Policy (Supervised Classification)

For fast reflex choices without re-running combinatorial planners, a Multi-Layer Perceptron (MLP) acts as a reactive policy:

- **Input Feature Vector ($\mathbf{x} \in \mathbb{R}^7$):** Normalized state features:

$$\mathbf{x} = [\text{loc_id}, \text{dist_goal}, \text{nearby_npcs}, \text{clue_flag}, \text{threat_level}, \text{caution}, \text{curiosity}] \tag{6}$$

- **Target Space ($y$):** Discrete action classes $y \in \{\text{Move}, \text{Talk}, \text{Examine}, \text{Wait}, \text{Flee}\}$.
- **Topology:** Input (7) $\rightarrow$ Dense (32, ReLU) $\rightarrow$ Dense (16, ReLU) $\rightarrow$ Softmax (5).
- **Training:** Optimized using Adam and cross-entropy loss over simulation logs; evaluated via held-out confusion matrices and F1-scores.

### 4.4 Unsupervised Trajectory Clustering (K-Means)

To study emergent playstyles without manual behavioral tagging, simulation trajectory summaries (total distance, conversational volume, clue inspections, idle ticks) are clustered using K-Means ($k = 3$):

1. **The Inquisitor:** High spatial displacement, frequent clue inspections, moderate dialogue.
2. **The Socialite:** Low transit distance, long stays at the Café and Town Square, high dialogue frequency.
3. **The Cautious Resident:** Avoids high-threat alleys, returns home early, minimizes risky investigations.

### 4.5 Grounded LLM Dialogue Engine

The LLM is invoked solely when linguistic exchange occurs. A controlled context-injection pipeline compiles the character's private memories and persona into a strict JSON contract:

```
{
  "system": "You are Ahmed, a 42-year-old cafe owner. Cautious and guarded with
     strangers. Answer strictly based on your private memories. If you did not
    witness an event, truthfully say you do not know.",
  "npc_profile": { "name": "Ahmed", "traits": ["cautious", "observant"], "
    current_goal": "Keep cafe safe" },
  "relevant_memories": [
    "[09:30 AM] Brewed coffee for morning patrons.",
    "[10:00 AM] Sat reading the town newspaper."
  ],
  "player_input": "Did you see who dropped the golden watch at the Market?",
  "output_format": "JSON: {\"reply\": string, \"mood\": string}"
}
```

Listing 1: Example Grounded LLM Prompt Template

When evaluated by the model, this structured context produces a grounded, in-character response without leaking unobserved town events:

```
{
  "reply": "A golden watch? I haven't left this cafe all morning, so I couldn't
     tell you. You might want to check down at the Market yourself.",
  "mood": "neutral"
}
```

Listing 2: Example Grounded LLM Response Output

Because Ahmed's prompt excludes events from the Market, he cannot leak information regarding the stolen watch, preserving narrative authenticity.

# 5 The Town Environment and Knowledge Boundaries

Believable multi-agent social behavior requires **information asymmetry**. Knowledge must diffuse across the environment through physical encounters rather than instantaneous global updates (Figure 4).

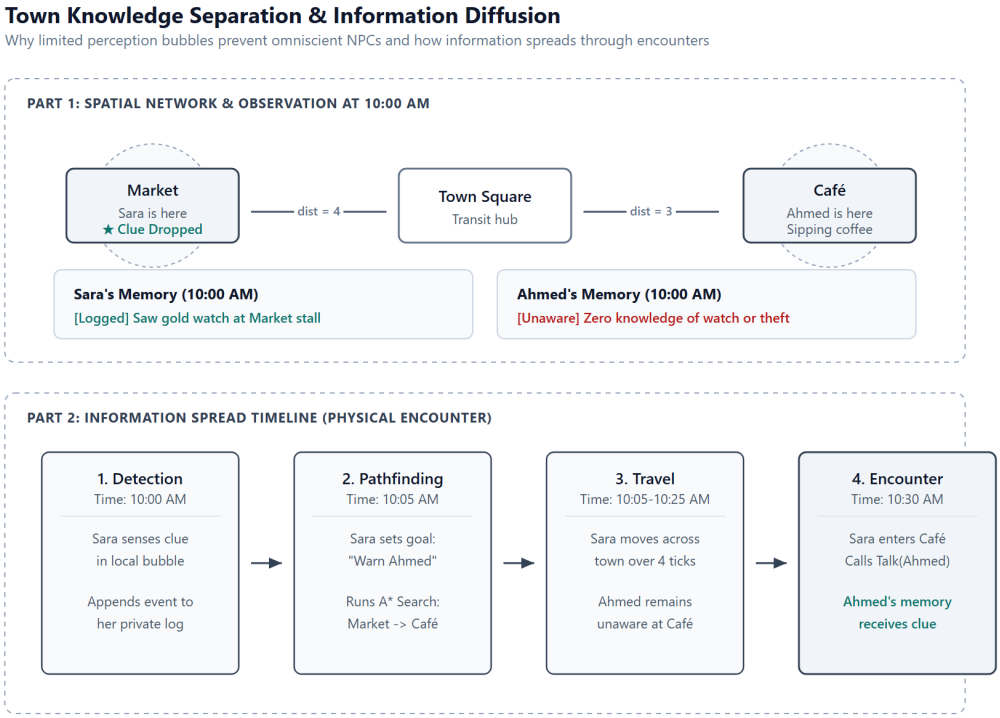



Figure 4: Spatial perception asymmetry at 10:00 AM and information diffusion timeline.

## 5.1 Information Diffusion Scenario Walkthrough

1. **Detection (10:00 AM):** A gold pocket watch is dropped at Market Stall 4. Sara is located at the Market node; her sensory filter logs the clue. Ahmed is at the Café node ($d = 7$ ticks away); his sensory radius detects nothing.

2. **Goal Formulation and Pathfinding (10:05 AM):** Sara establishes a goal: "Warn Ahmed at Café." She runs $A^*$ search to compute the route: Market $\rightarrow$ Town Square $\rightarrow$ Café.

3. **Physical Transit (10:05–10:25 AM):** Sara traverses intermediate nodes over multiple clock ticks. Ahmed remains reading at the café, completely oblivious.

4. **Social Encounter (10:30 AM):** Sara enters the Café node and executes Talk(Ahmed). The dialogue model generates speech: *"Ahmed, look what was dropped at the Market!"* Ahmed's memory ledger appends: `[10:30 AM] Sara reported finding a gold watch at Market.`

# 6 Data Generation and Machine Learning Pipeline

The project implements a self-generating simulation pipeline on CPU hardware, avoiding massive web datasets or GPU clusters (Table 2).

Table 2: Self-Generating Simulation Data Pipeline Specifications

| Pipeline Stage | Technical Implementation |
|---|---|
| **1. Simulation Logging** | Execute 1,000–5,000 automated episodes (20–50 ticks each) with heuristic agents. Record state-action transitions to SQLite/JSON. |
| **2. Feature Extraction** | Convert turns into 7-dimensional normalized vectors $\mathbf{x}$ and discrete action labels $y \in \{\text{Move}, \text{Talk}, \text{Examine}, \text{Wait}, \text{Flee}\}$. |
| **3. Model Training** | Split dataset 80/20. Train MLP classifier in `scikit-learn`. Run K-Means ($k = 3$) on agent trajectory matrices. |

Generating 2,000 automated episodes and training the policy classifier completes in under 5 minutes on a modern laptop CPU.

# 7 Implementation Plan and 8-Week Roadmap

The project schedule is structured across eight weeks for a two-student team (Table 3).

Table 3: Eight-Week Phased Project Implementation Schedule

| Period | Milestone Focus | Team Responsibilities |
|---|---|---|
| **Weeks 1–2** | World Engine & NPC Schema | Student A: Graph topology & tick loop engine.<br>Student B: NPC state & perception schemas. |
| **Weeks 3–4** | Classical AI Algorithms | Student A: $A^*$ search & graph traversal.<br>Student B: GA chromosome encoding & fitness. |
| **Week 5** | LLM Dialogue & UI | Student A: FastAPI REST endpoints & web map.<br>Student B: Prompt templates & JSON validation. |
| **Week 6** | Machine Learning Pipeline | Student A: Automated data generation scripts.<br>Student B: MLP training & K-Means clustering. |
| **Week 7** | Evaluation & Ablations | Both Students: 100 scenario trials, consistency benchmarks, and demographic bias tests. |
| **Week 8** | Final Deliverables | Both Students: Report authoring, chart synthesis, code packaging, and demonstration video. |

## 7.1 Difficulty Assessment ($\sim 7/10$)

- **Simulator Engine (5/10):** Standard Python graph programming using NetworkX and Gymnasium conventions.
- **Genetic Algorithm (6/10):** Straightforward permutation encoding; requires tuning objective weights ($w_1 \dots w_4$).

- **MLP and Clustering (6/10):** Standard scikit-learn workflow; CPU training executes rapidly.
- **Grounded Consistency (8/10):** Preventing LLM factual leaks and hallucinations requires disciplined JSON prompt engineering and rigorous validation gates.

# 8 Quantitative Evaluation Framework

The system's performance is rigorously measured across 100 scenario trials:

## 8.1 Core Evaluation Metrics

1. **Goal Success Rate:** Percentage of episodes where NPCs fulfill assigned primary goals within 50 ticks (Target: $\geq 80\%$).

$$\text{Success Rate} = \frac{N_{\text{completed}}}{N_{\text{total}}} \times 100\% \tag{7}$$

2. **Path Optimality Ratio:** Ratio of optimal $A^*$ path length to actual traversed distance (Target: $\geq 92\%$).

$$\text{Optimality} = \frac{\text{Length}(A^*)}{\text{Distance}_{\text{traveled}}} \tag{8}$$

3. **Memory Consistency Rate:** Automated verification that conversational statements contain zero contradictions against memory logs (Target: $\geq 95\%$).

$$\text{Consistency} = 1 - \frac{N_{\text{contradictions}}}{N_{\text{dialogues}}} \tag{9}$$

4. **Goal Alignment Score:** Proportion of actions directly advancing top-priority goals versus idle ticks (Target: $\geq 85\%$).

$$\text{Alignment} = \frac{A_{\text{goal-relevant}}}{A_{\text{total actions}}} \tag{10}$$

5. **ANN Classification F1-Score:** Weighted F1-score of the MLP policy on held-out test data (Target: $\geq 0.88$).

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

## 8.2 Ablation Study Matrix

To evaluate the contribution of each component, four configurations are compared (Table 4).

Table 4: System Ablation Experimental Comparison Matrix

| Configuration | Decision Engine | Memory | Observed Limitation |
|---|---|---|---|
| **1. Full Hybrid (Ours)** | $A^*$ + GA + MLP | Enabled | Optimal performance baseline. |
| **2. Amnesic Baseline** | $A^*$ + GA + MLP | Disabled | Severe dialogue repetition. |
| **3. Rule-Based Baseline** | Static State Machine | Enabled | Rigid, zero conversational nuance. |
| **4. Pure LLM Baseline** | Prompting Only | Enabled | Frequent physical hallucinations. |

### 8.3 Demographic Bias and Fairness Testing

Following benchmark conventions (e.g., BBQ), mystery scenarios are executed while systematically altering character names and demographic markers (e.g., Ahmed vs. Sara vs. Alex). Statistical significance tests confirm that dialogue responses exhibit invariance in sentiment, suspicion, and blame attribution.

## 9 Summary and Key Advantages

1. **Pragmatic Resource Footprint:** Runs 100% on consumer-grade laptop CPUs without requiring expensive cloud GPUs or model fine-tuning.
2. **Comprehensive AI Coverage:** Unifies classical search ($A^*$), evolutionary computation (GA), supervised learning (MLP), unsupervised clustering (K-Means), and grounded generative AI.
3. **Verifiable Grounding:** Restricting the language model to dialogue while anchoring physical reality in deterministic simulation code guarantees that generative hallucinations cannot violate game physics.

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